Tyler Gaines

tylergaines05@gmail.com | 832-221-3453 | LinkedIn | tytaniumg.github.io

Professional Summary

Computer Science student with a practical, efficiency-first approach to problem-solving and a strong foundation in systems, algorithms, and software design. Experienced in building and benchmarking complex data-driven systems, from custom data structures to full-stack web projects. Seeking a CS internship to apply technical skills in a collaborative, professional environment.

Education

The University of Texas at Dallas Richardson, TX

Bachelor of Science, Computer Science. *Expected May 2028*

Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Systems Programming

Certifications

- CompTIA Security+, September 2026

Skills

Programming Languages: C, C++, C#, Java, Python, Assembly, HTML, CSS

Operating Systems: Linux, UNIX, Windows

Tools: Bash, Git, GitHub, VS Code, MS Office

Languages: English, Portuguese

Experience

Ride Attendant / Lifeguard June 2026 – August 2026

Schlitterbahn *Galveston, TX*

- Operated water attractions efficiently while enforcing strict safety procedures and ensuring a positive guest experience.
- Assisted guests with questions, directions, ride procedures, and general park needs.
- Monitored pools and ride areas for signs of distress, injury, or drowning and responded quickly to emergencies.
- Performed water rescues and provided first aid when necessary, prioritizing guest safety.

Projects

In-Memory Document Search Engine January 2026 – May 2026

Java | Data Structures and Algorithms Course

- Independently designed a multi-index document store with a custom circular buffer, TreeMap, and hash-based inverted index, enabling O(1) average-case term lookup and O(log n) prefix/range queries.
- Implemented multi-term AND search using a size-ordered set-intersection strategy to minimize computation on large posting lists.
- Benchmarked performance across 10K–100K document datasets, achieving ~1.5µs average term search latency and validating theoretical complexity against measured results.
- Maintained consistency across four synchronized data structures through all insert/delete operations.

Personal Portfolio Website https://tytaniumg.github.io/ October 2026

HTML, CSS, Git, GitHub Pages

- Designed and built a responsive portfolio site from scratch with semantic HTML5 and CSS (Flexbox, custom properties), tested on desktop and mobile screen sizes.
- Added accessibility features, including descriptive aria-label text on icon links and keyboard-focus styling on interactive elements.
- Deployed with GitHub Pages using Git version control, maintained through ongoing commits and pushes.

Activities & Leadership

Cyber Defense and Response Unit August 2025 – May 2026

Member

Computer Security Group August 2024 – May 2025

Member